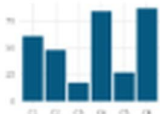
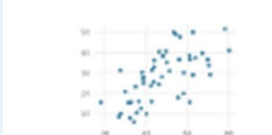
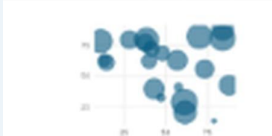
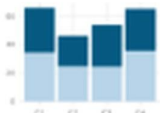
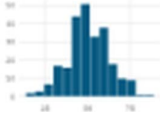


Visualization Matching Game - Answers

Name: Udhayakumar U

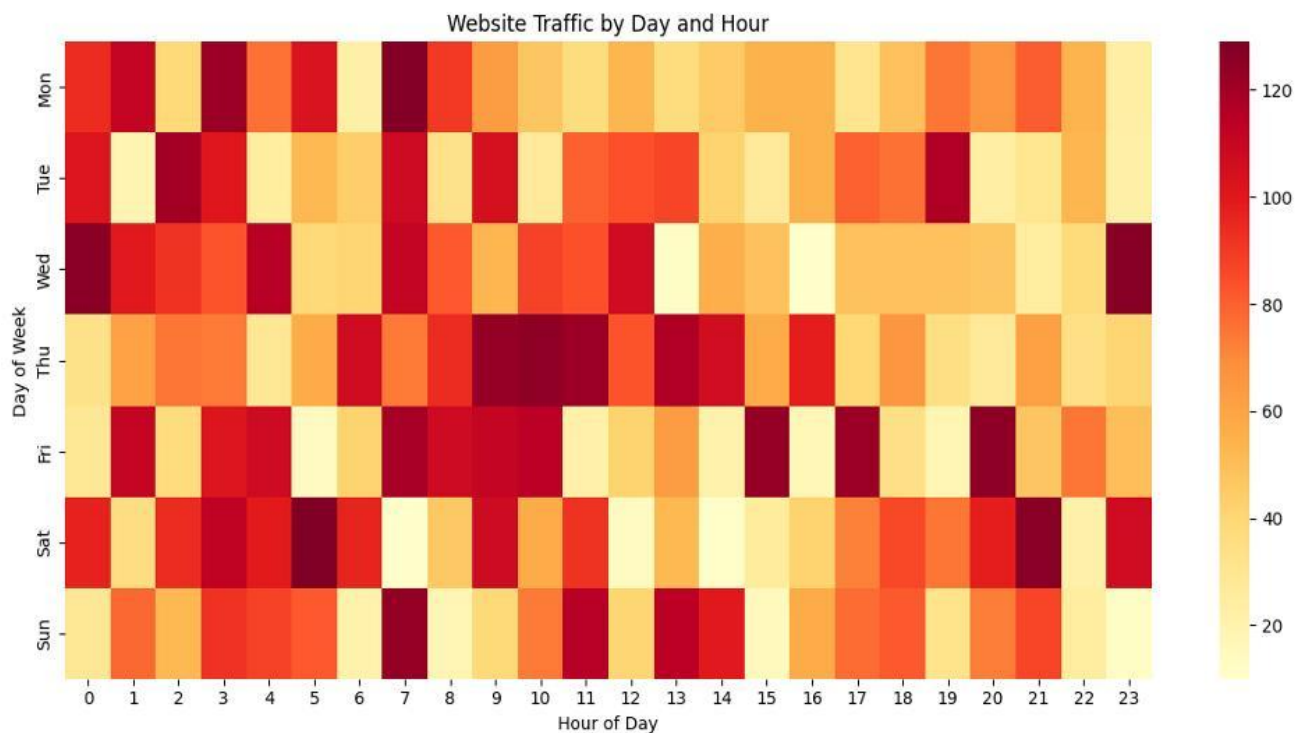
Reg. No: 192424332

Course Code: DSA 0601 (Data Handling and Visualization)

Q.No	Question	Plot	Answer
1	Shows magnitude across two categories using color intensity		A Heatmap
2	Shows how a value changes over time		I Line chart
3	Compares amounts across separate categories		C Bar chart
4	Shows the relationship between two continuous variables		E Scatter plot
5	Shows how parts contribute to a whole		D Pie chart
6	Shows three variables at once using position and size		F Bubble chart
7	Shows part-to-whole composition across categories		B Stacked bar chart
8	Compares distributions across groups and highlights outliers		G Box plot
9	Shows the distribution of a single continuous variable		H Histogram
10	Shows how a cumulative total changes over time		J Area chart

1. Heatmap – A

- A heatmap uses **color intensity** to represent values.
- Darker or stronger colors usually mean **higher values**.
- Lighter colors represent **lower values**.
- It is useful for comparing values across **two categories or dimensions**.
- **Example:** Student marks by subject and month.



2. Line Chart – I

- A line chart shows **how a value changes over time**.
- Data points are connected with lines.
- It is useful for identifying **increases, decreases, and trends**.
- **Example:** Monthly sales from January to December.

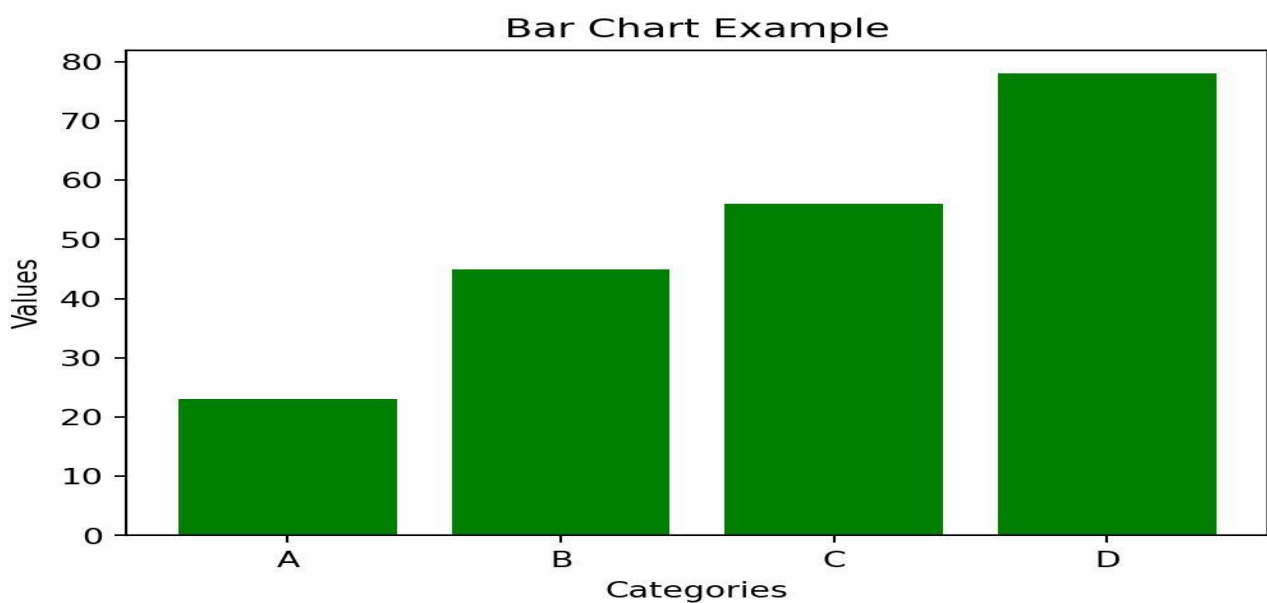
LINE CHART

Total product quantity sold for each month



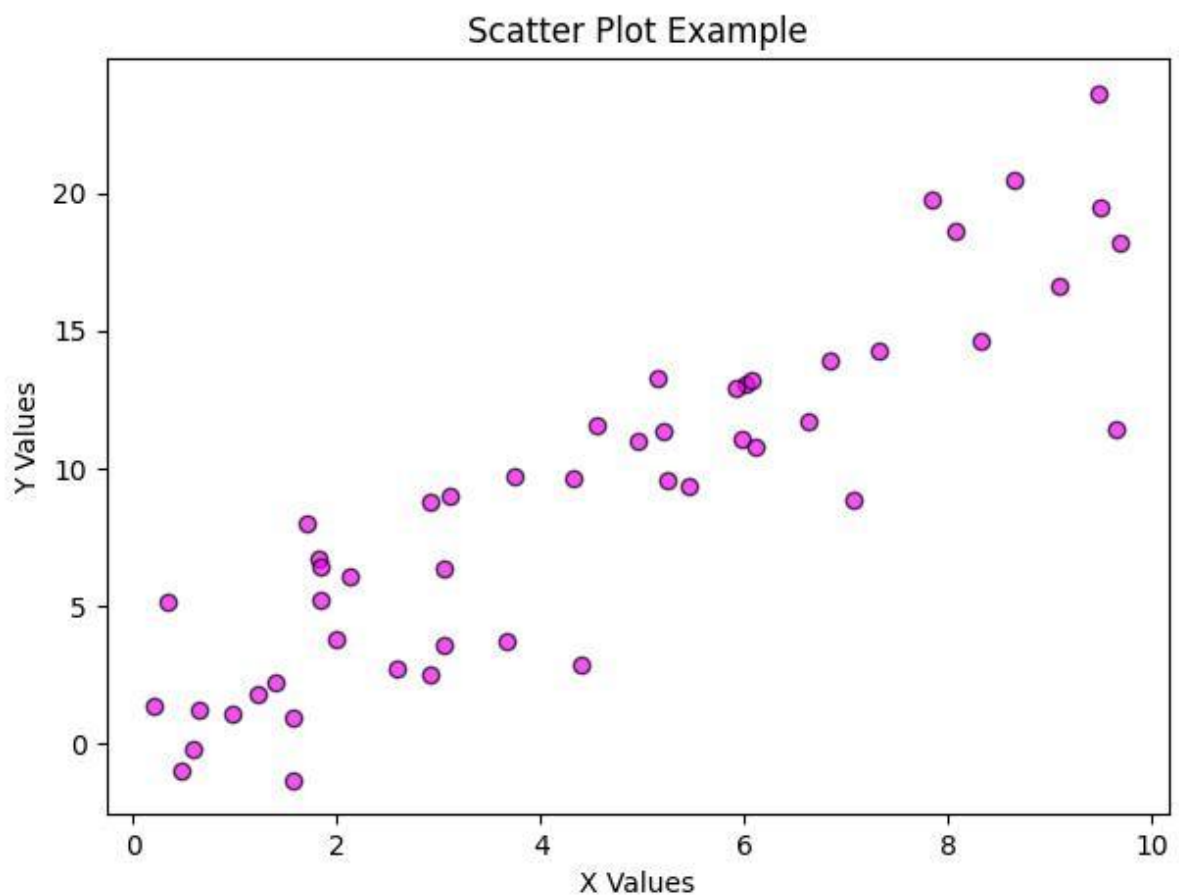
3. Bar Chart – C

- A bar chart compares **different categories**.
- The length or height of each bar represents the value.
- It is useful for easy comparison between categories.
- **Example:** Number of students in different departments.



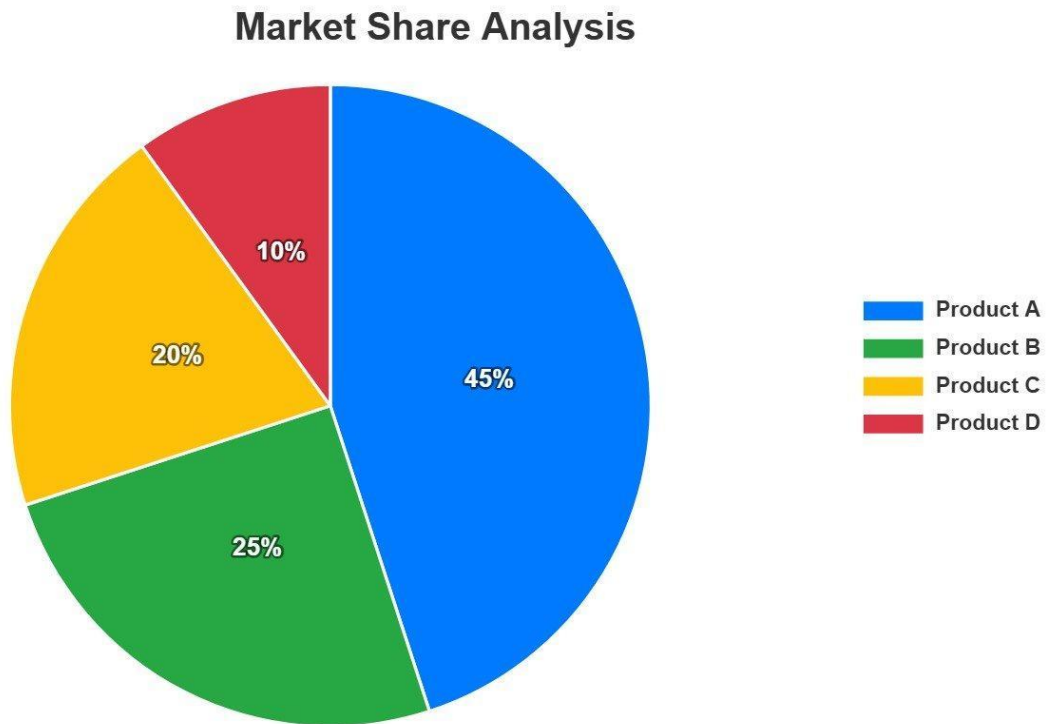
4. Scatter Plot – E

- A scatter plot shows the **relationship between two continuous variables**.
- Each dot represents one observation.
- It helps identify **correlation, patterns, or unusual values**.
- **Example:** Height vs. weight.



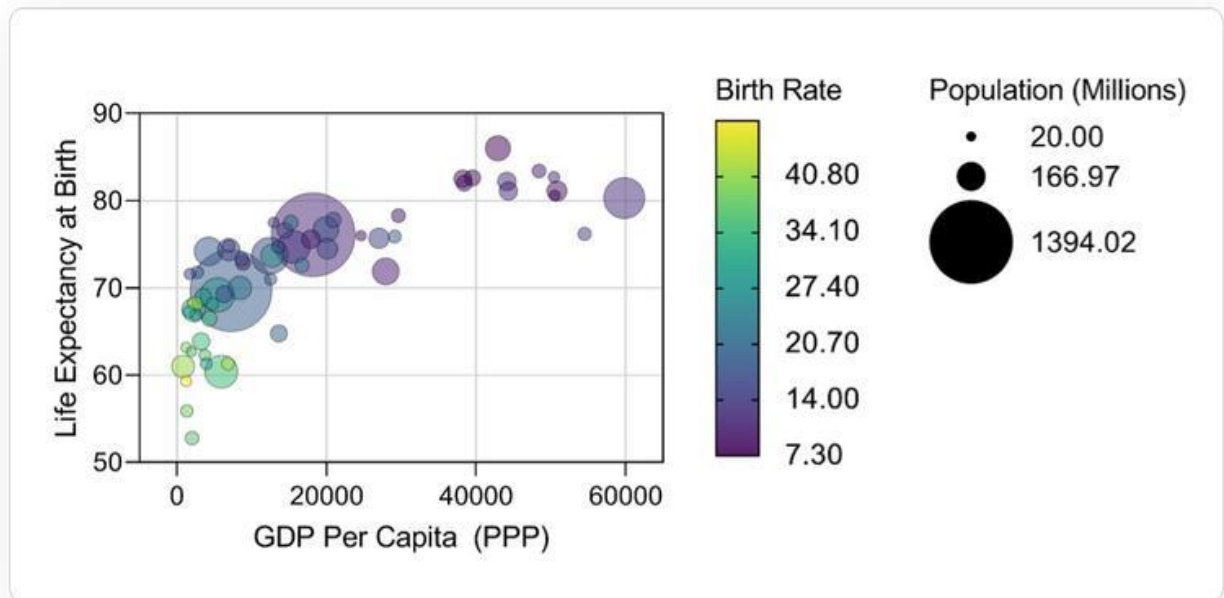
5. Pie Chart – D

- A pie chart shows how different parts contribute to a **whole**.
- The complete circle represents **100%**.
- Each slice represents a category's share.
- **Example:** Percentage of students in different departments.



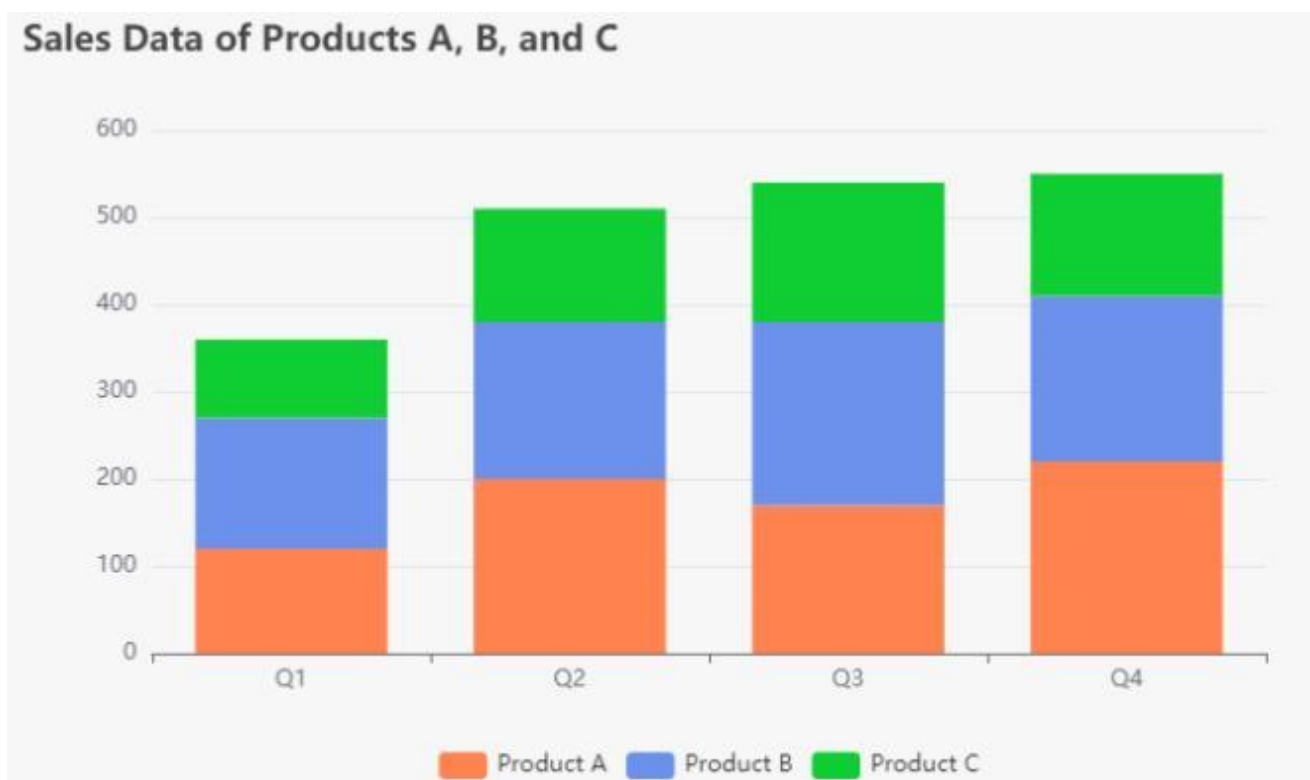
6. Bubble Chart – F

- A bubble chart displays **three variables at once**.
- The **x-position** represents one variable.
- The **y-position** represents another variable.
- The **bubble size** represents a third variable.
- **Example:** Sales, profit, and number of products.



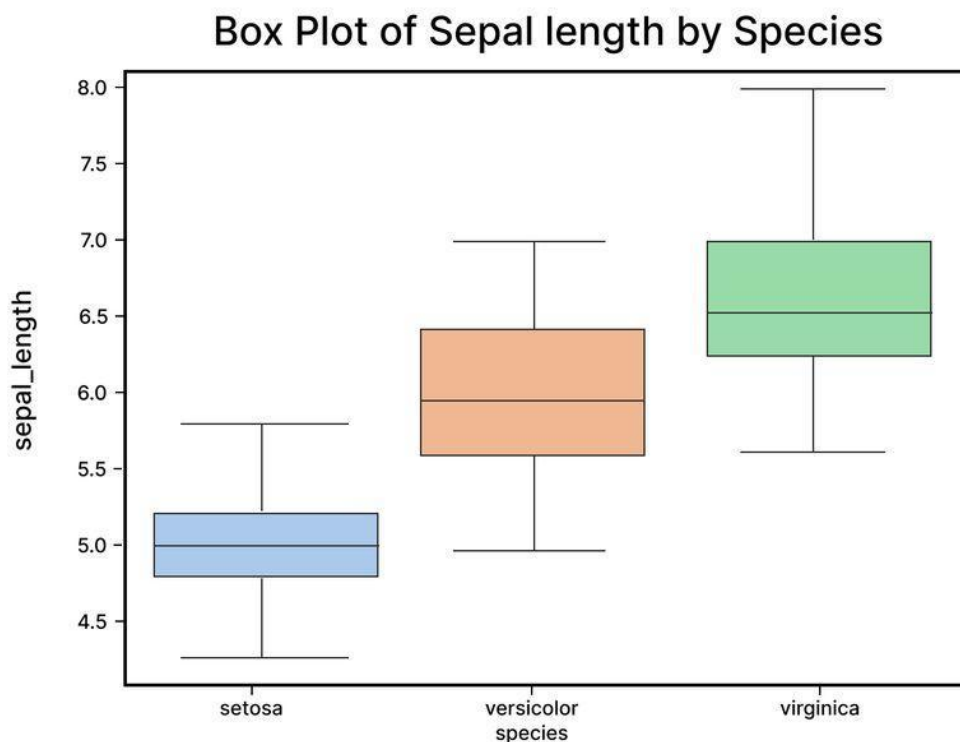
7. Stacked Bar Chart – B

- A stacked bar chart shows **part-to-whole composition across categories**.
- Each bar is divided into different sections.
- Each section represents a part of the total.
- **Example:** Approved and rejected applications for different groups.



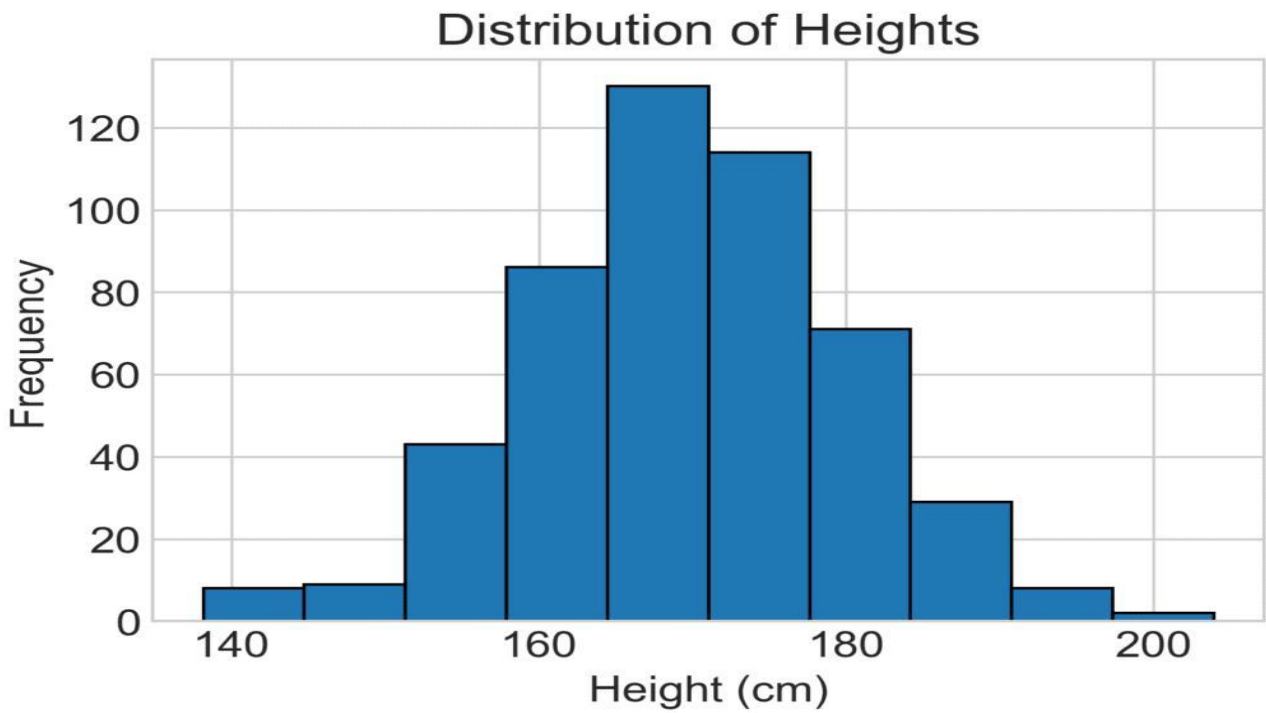
8. Box Plot – G

- A box plot compares **distributions between groups**.
- It shows the **median, quartiles, spread, and outliers**.
- It is useful for identifying unusual values.
- **Example:** Exam-score distribution across different classes.



9. Histogram – H

- A histogram shows the **distribution of one continuous variable**.
- Values are grouped into intervals called **bins**.
- It helps understand whether data is concentrated, spread out, or skewed.
- **Example:** Distribution of students' exam marks.



10. Area Chart – J

- An area chart shows how a value changes over **time**, with the area below the line filled.
- It is useful for showing **cumulative or overall magnitude**.
- **Example:** Total revenue accumulated over several months.

